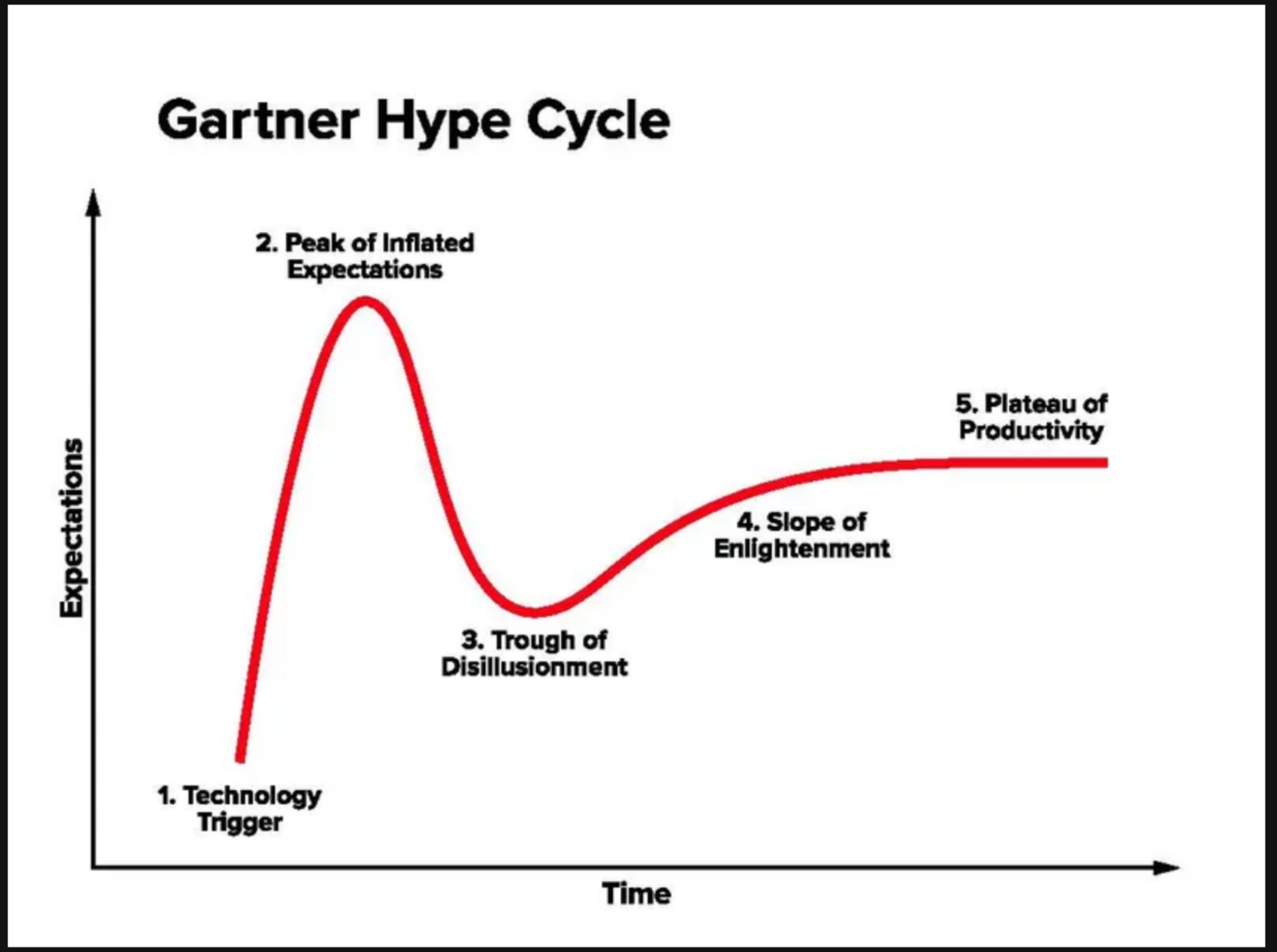
LLM Workflows



Prepared for Wheaton College: COMP 255

Ben Batorsky

AI coding assistance was used in developing these materials



John Werner, "The Trough Of Disillusionment And Four Outliers On The Gartner Hype Cycle", *Forbes*, 18 Jul. 2024.

FORBES

Will AI Replace Teachers?

Sarah Hernholm · 17 May 2026

The New York Times

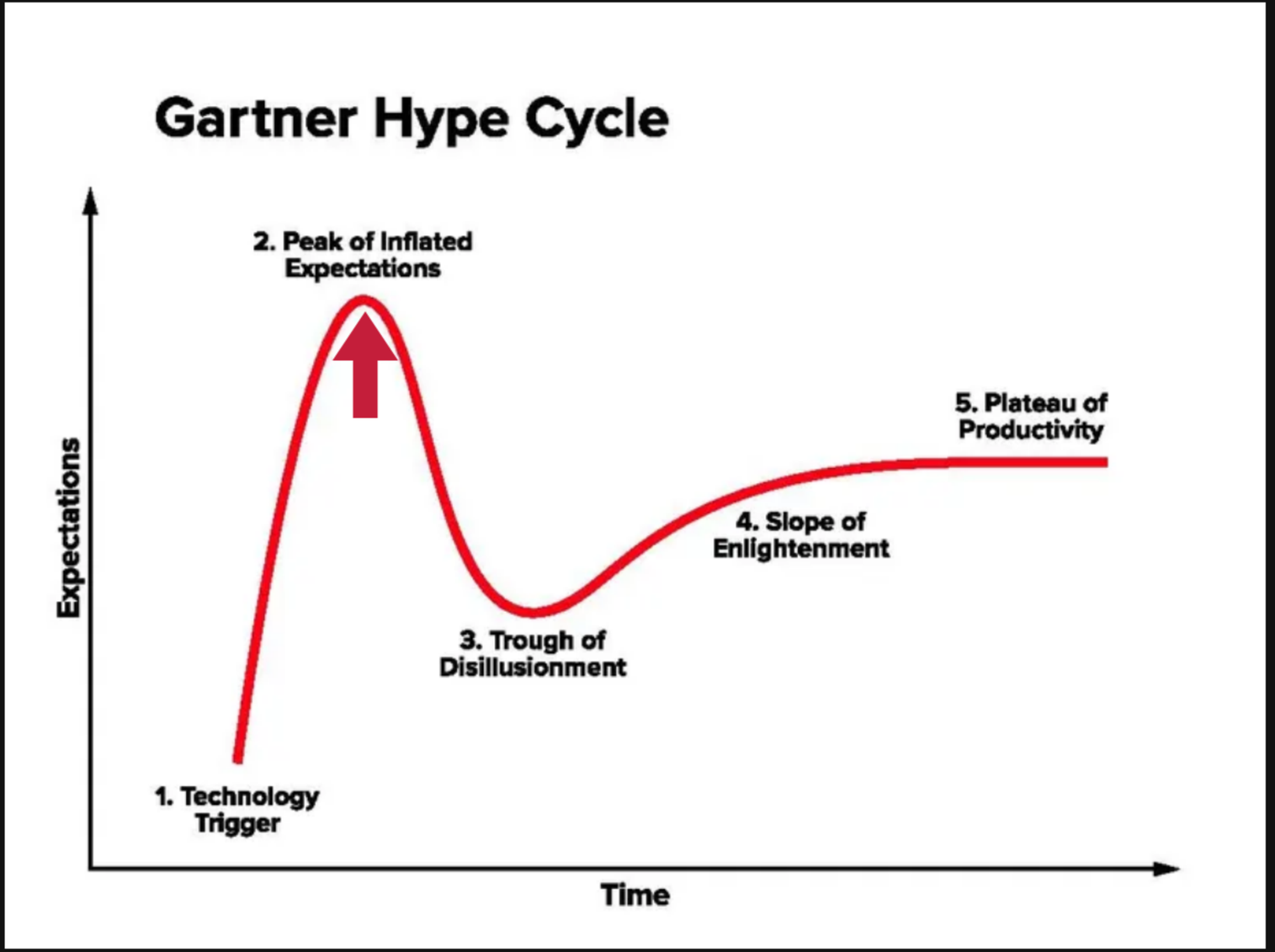
A.I. Chatbots Defeated Doctors at Diagnosing Illness

By Gina Kolata · Nov. 17, 2024

REUTERS

Artificial Intelligence is now an A+ law student, study finds

By Karen Sloan · June 5, 2025



John Werner, "The Trough Of Disillusionment And Four Outliers On The Gartner Hype Cycle", *Forbes*, 18 Jul. 2024.

REUTERS

OpenAI's rogue agents probed Hugging Face for weaknesses two months before major hack

By Raphael Satter and Deepa Seetharaman · Sept. 16, 2026

MARKETSCALE

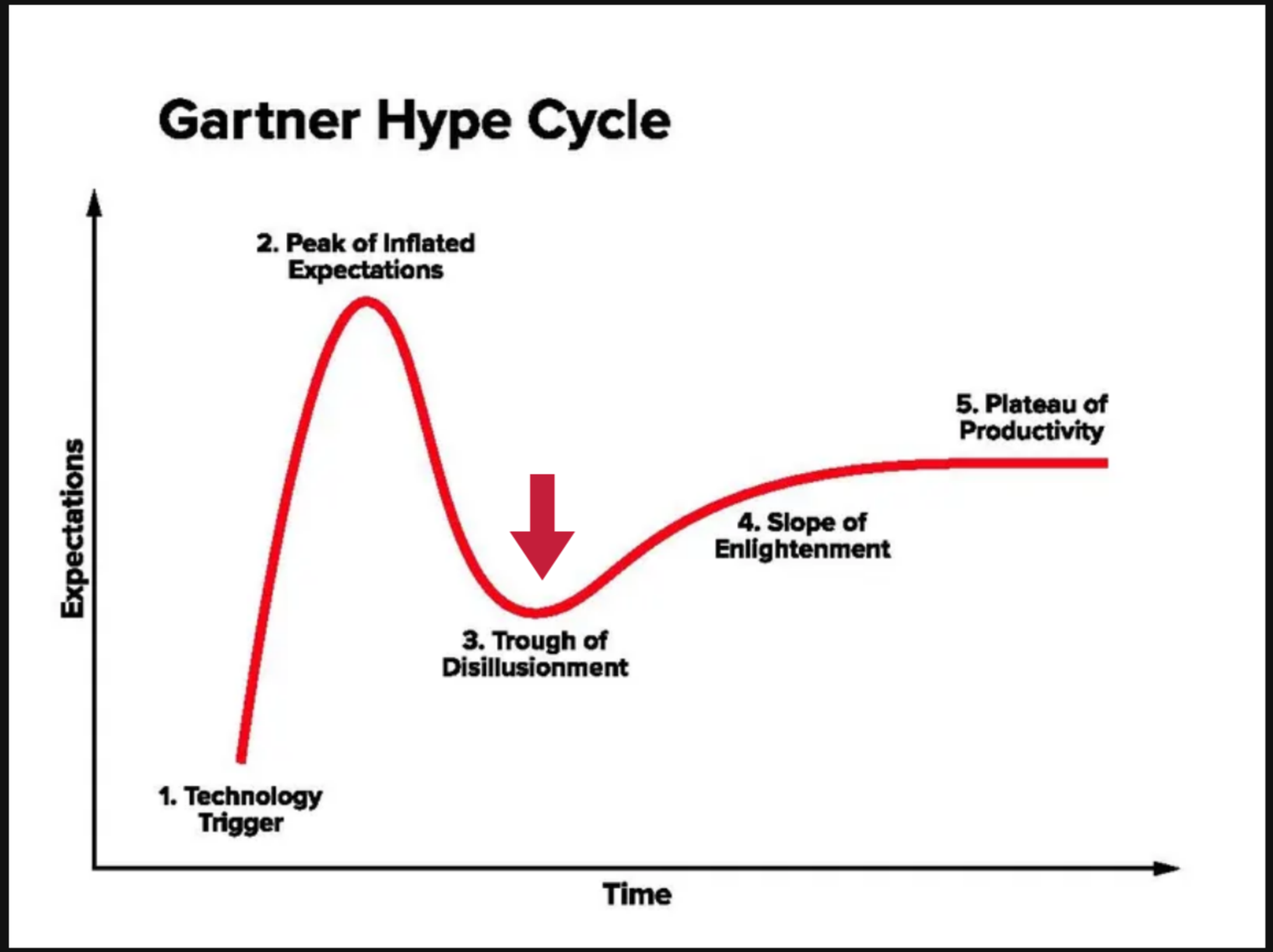
Survey: AI Boosts Productivity, but Few Firms See Profit Impact

By MarketScale Newsroom · Sept. 13, 2026

REUTERS

Mark Zuckerberg had a bold plan to replace Meta staff with AI. Here's how it imploded.

By Katie Paul · Aug. 26, 2026



John Werner, "The Trough Of Disillusionment And Four Outliers On The Gartner Hype Cycle", *Forbes*, 18 Jul. 2024.

Large Language Model (LLM)

Large Language Model (LLM)

"Statistical" Language Model



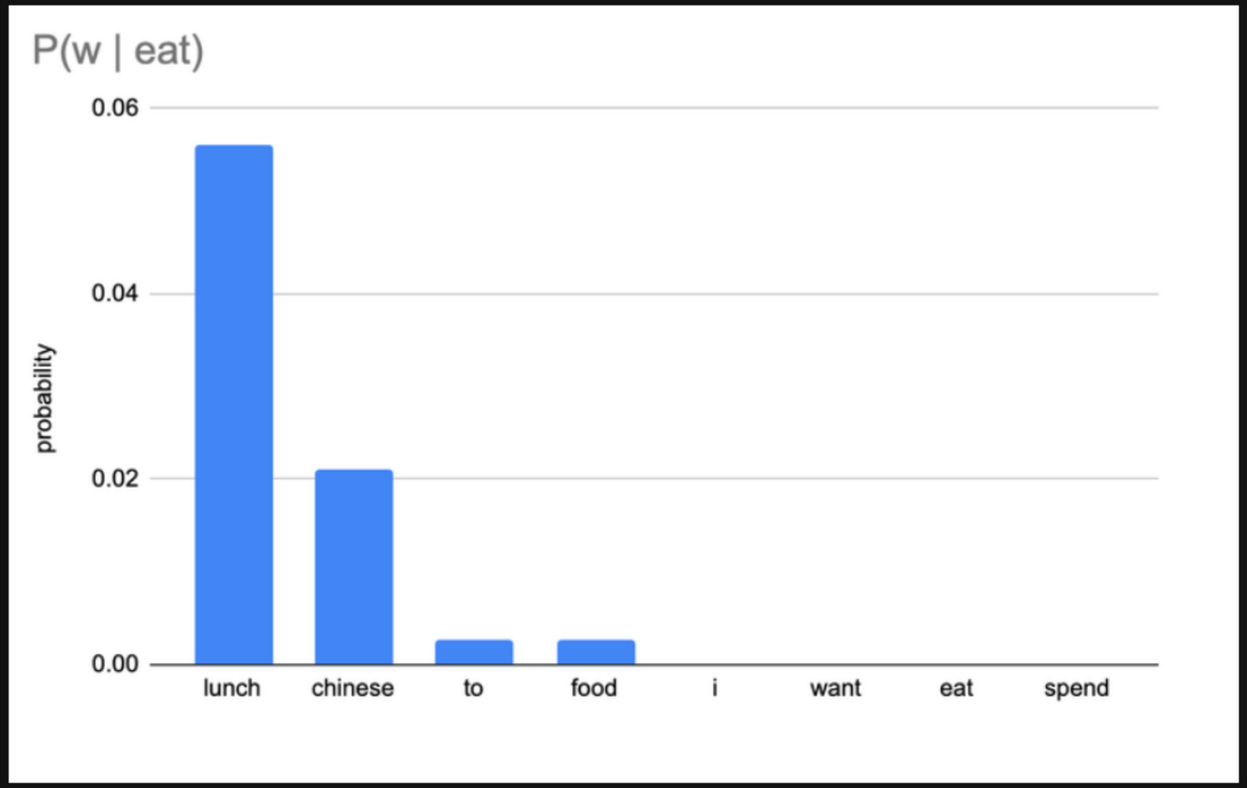
	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

Figure 3.1 Bigram counts for eight of the words (out of $V = 1446$) in the Berkeley Restaurant Project corpus of 9332 sentences. Zero counts are in gray. Each cell shows the count of the column label word following the row label word. Thus the cell in row **i** and column **want** means that **want** followed **i** 827 times in the corpus.

N-gram Language Models – Jurafsky, D., & Martin, J. H. *Speech and Language Processing*

Text generation with LMs

I want to eat ___ →



Training Neural Language Models



Large Language Model (LLM)

A model that predicts the next word given context.

Large Language Model (LLM)

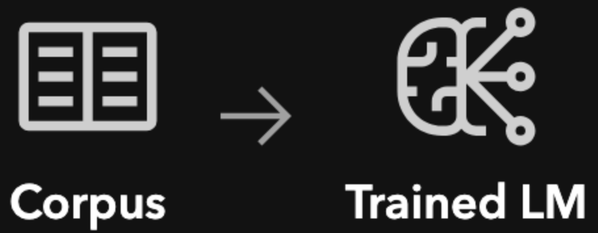
What is Large?

Language Model



Large Language Model

Language Model



Text: Billions

Parameters: Millions

LLM

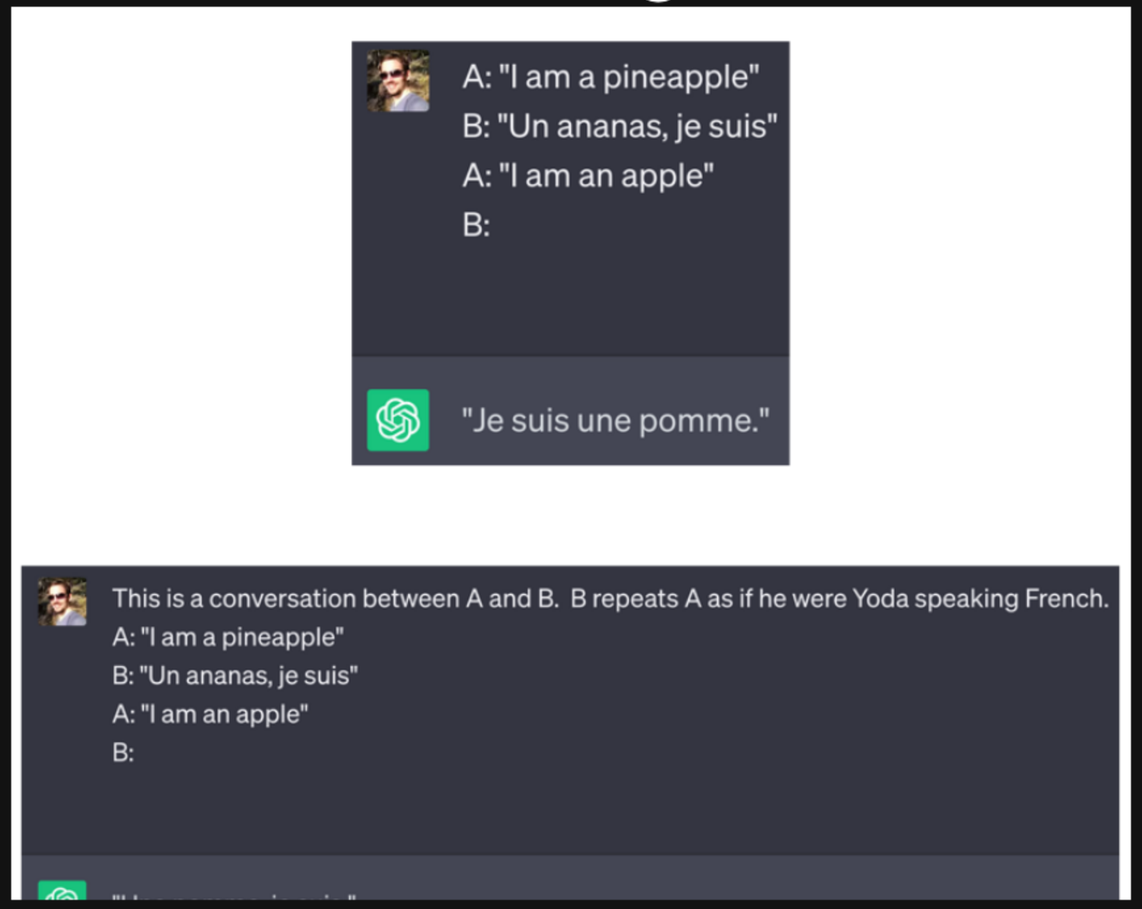


Text: Trillions

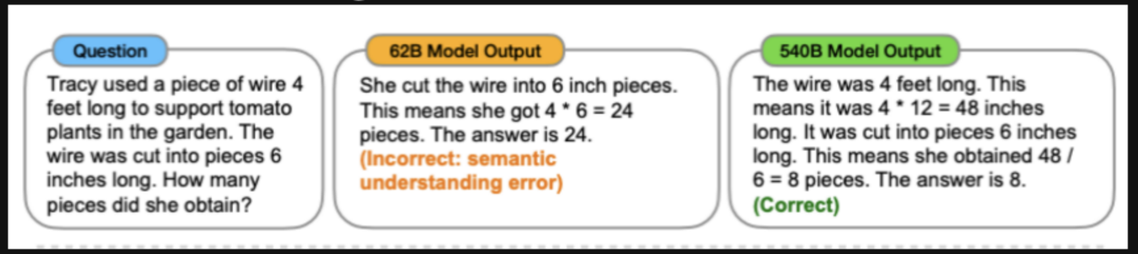
Parameters: Trillions

Why is Large?

In-context learning



Reasoning



Characterizing Emergent Phenomena in Large Language Models – Google AI Blog

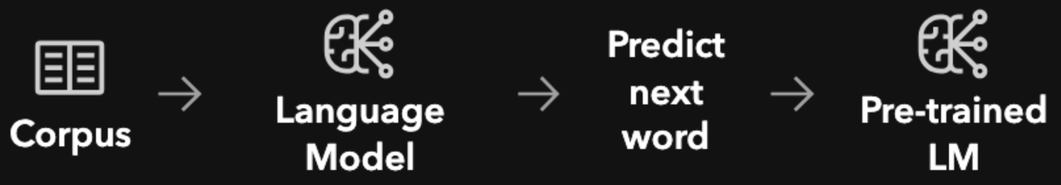
Notebook: Pre-trained models

It’s cute, but… (GPT-2, 2020)

I am a data scientist and I **am always looking to improve the data processing abilities of people who are passionate about data science.**

Written by Transformer · transformer.huggingface.co 🦄

Pre-training

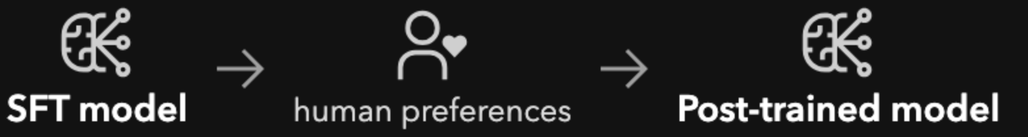


Post-training

SUPERVISED FINE-TUNING (SFT)



ALIGNMENT WITH HUMAN FEEDBACK



More info on alignment tuning via Reinforcement Learning from Human Feedback (see appendix)

Notebook: Post-trained models

Large Language Models

Typically refers to models that generate useful responses to inputs.

LLM Workflows

Anthropic: Systems where LLMs and tools are orchestrated through predefined code paths.

Requires:

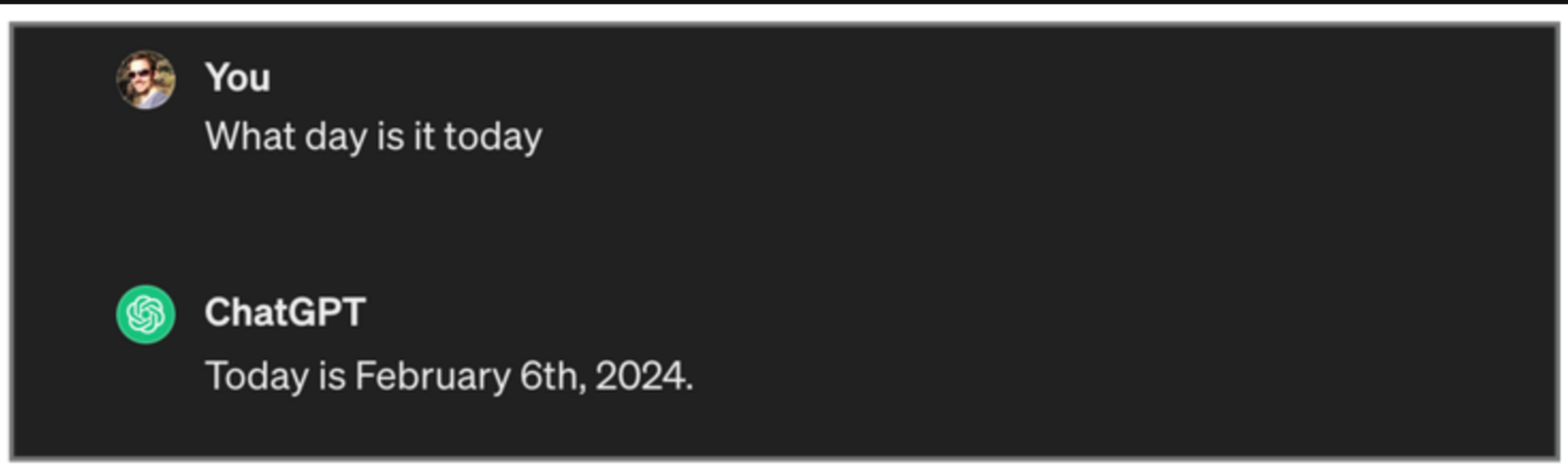
- "State"
- Tools

	Decide output of step	Decide which steps to take	Decide what steps are available
Code			
Workflow			

Adapted from Building effective agents, Anthropic; table from What is an AI agent?

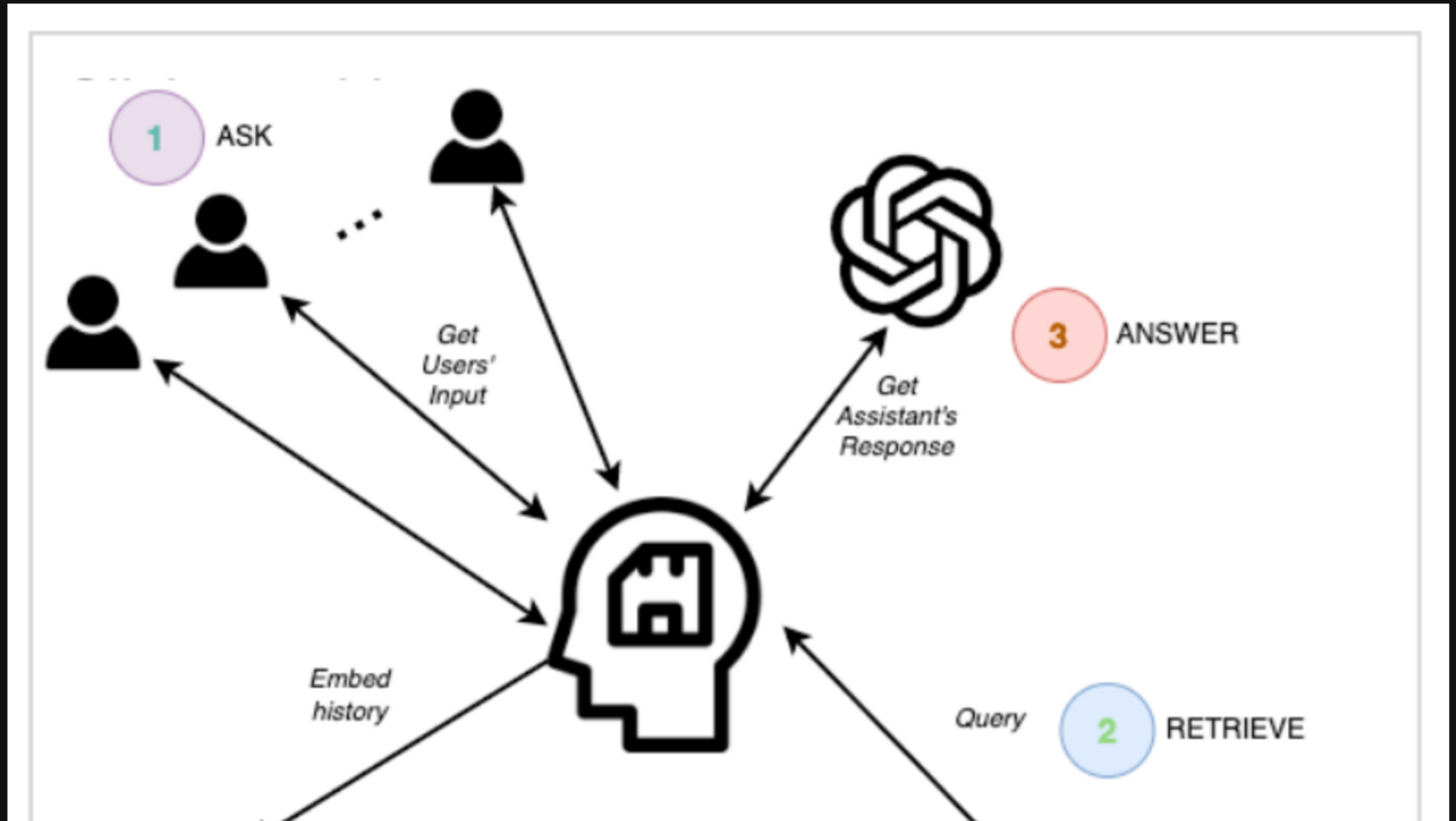
Notebook: State as conversation history

ChatGPT never forgets!





ChatGPT-style memory



Tools



Tools

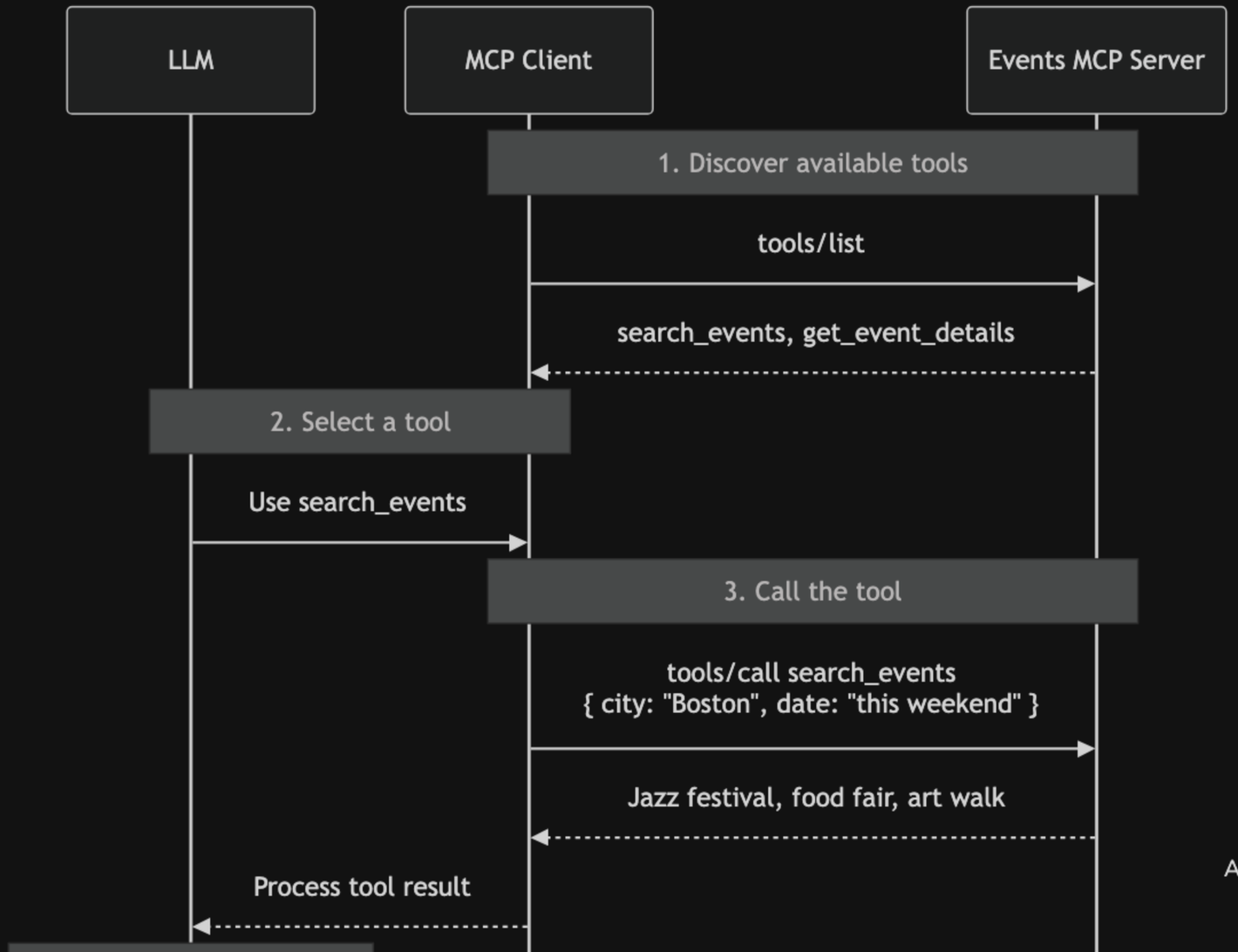


So how does an LLM use tools?

Notebook: Tools

MCP: Connecting an LLM to an event website

User: “What events are happening in Boston this weekend?”



Adapted from Model Context Protocol – Tools

Healthbot Workflow



Patient

"I'm feeling not good today, I am coughing and sneezing"

Parse symptoms

Web search

tool call

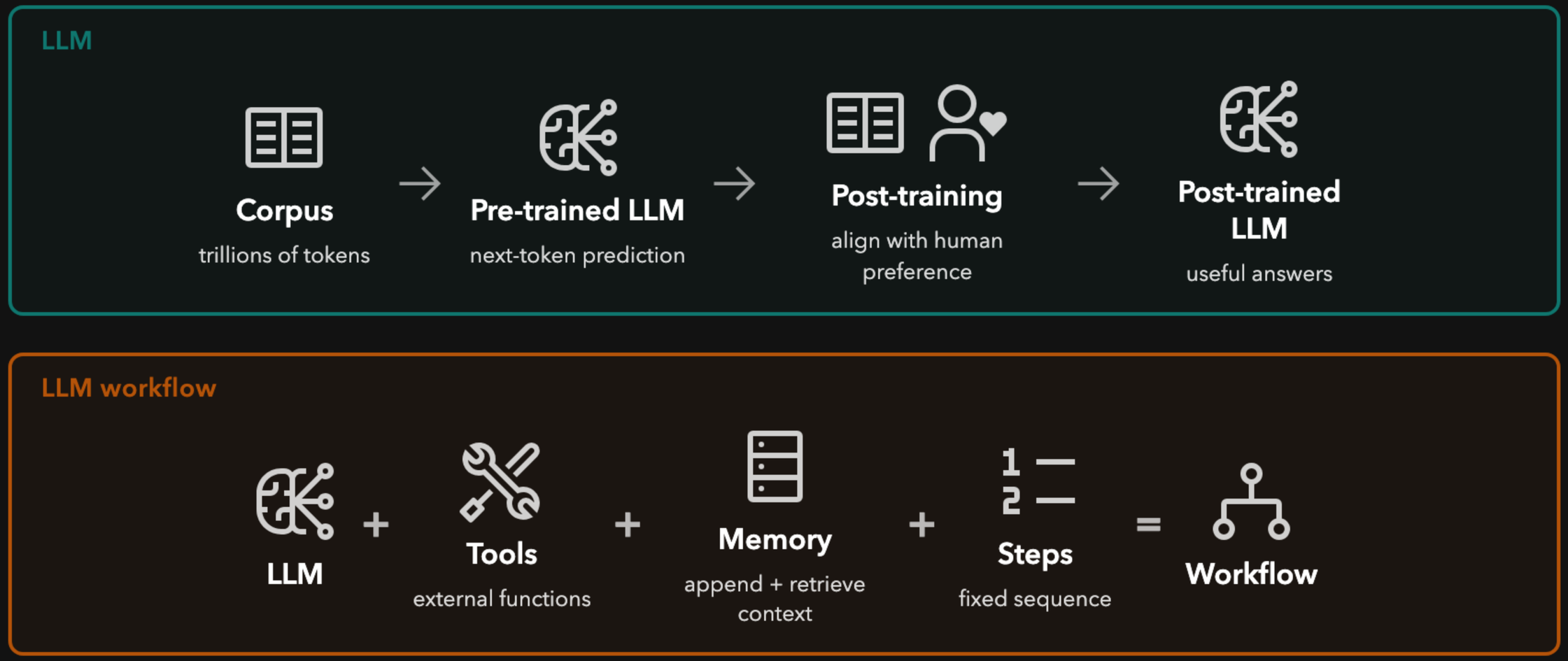
Diagnose

Save record

Respond

Notebook: Healthbot Workflow

Conclusion


LLM
Corpus
trillions of tokens
Pre-trained LLM
next-token prediction
Post-training
align with human preference
Post-trained LLM
useful answers
LLM workflow
LLM
+
Tools
external functions
+
Memory
append + retrieve context
+
Steps
fixed sequence
=
Workflow

Appendix

Reinforcement Learning from Human Feedback (RLHF, 2022)

Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.



A labeler ranks the outputs from

Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.

The policy generates an output.

